Describing the Structure of Dynamical Systems

A Unifying Mathematical Framework

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Abstract

A wide class of physical laws share a common structure: observable behavior emerges from the joint action of an intrinsic response and an extrinsic drive. Ohmic conduction, Fourier heat conduction, Fick diffusion, linear elasticity, Newtonian mechanics, and linear response theory (Onsager 1931) all admit representations of the form

$$\text{Behavior} = \text{Response} \times \text{Drive}$$

In the simplest (linear, memoryless) cases, the interaction is literal multiplication — a material coefficient times a gradient or driving variable. More generally, "Response × Drive" denotes the action of an operator on a state: a bilinear pairing, a convolution, or a local linearization. The product should be read as a structural motif, not exclusively as arithmetic multiplication. At the field level, constitutive relations combine with conservation laws to yield evolution equations of the form

$$\frac{\partial u}{\partial t} = \mathcal{L}_\theta[u] + S(x, t)$$

where $\mathcal{L}_\theta$ is an operator determined by intrinsic structure (diffusivity, conductivity, stiffness, permittivity) and S represents extrinsic sources and boundary drives. This defines an ordinary differential equation on an infinite-dimensional manifold of field configurations whose geometry — modes, curvature, stable and unstable directions — is largely set by intrinsic properties, whereas realized system histories are selected by extrinsic drives and initial conditions.

This interaction structure has profound epistemological consequences. Because the observable is generated by the joint action of intrinsic and extrinsic factors, behavior alone cannot uniquely resolve "how much was the response" versus "how much was the drive" without auxiliary assumptions; causal attributions are **structurally underdetermined** along a nature–nurture axis. The individual ingredients of this framework — constitutive laws and linear response theory (Onsager 1931), bond-graph power conjugacy (Paynter 1961; Karnopp, Margolis, and Rosenberg 2012), evolution equations on function spaces (Pazy 1983), and identifiability analysis — are each well established. What this work contributes is a **unifying narrative**: a continuous logical arc from the constitutive-law observation through cross-domain analogies, operator-form PDEs, and control theory to a formal demonstration that the interaction structure implies structural underdetermination of causal attributions at every level. We also identify explicit limits of validity — where the framework breaks (chaos, strong nonlinearity, memory effects) and where it remains predictive.

1 Part I: The Primordial Pattern

1.1 Why Change Matters

All physical laws describe change (or its absence). Dynamical systems provide a rigorous mathematical framework for these descriptions. The central question: what encourages or discourages a system to change? Is the influence intrinsic to the system or imposed by its environment? Perhaps both? And in what proportion?

1.2 Two Laws, One Pattern

Of key significance is the classical relationship between mass, acceleration, and force—described by Newton's Second Law—and the relationship between charge, the electric field, and the electrostatic force—given by Coulomb's Law:

Newton's Second Law:

$$F = ma$$

Coulomb's Law:

$$F = qE$$

Note both have similar forms, each denoting a force as the product of two interacting components—a *source* (mass m, or charge q) and a *field* (acceleration a, or electric field E). Interestingly, one can combine the words "source" and "field" to create the word "force"—a useful mnemonic. The generalized force can be expressed as:

$$\boxed{\text{Force} = \text{Source} \times \text{Field}}$$

Force is measured in Newtons in both cases. Mass is measured in kilograms and charge is measured in Coulombs. Thus, the units of acceleration are Newtons per kilogram (m/s^2), and the units of the electric field are Newtons per Coulomb (V/m). If the electric field is considered analogous to an acceleration field, then the unit for generalized field strength is:

$$\text{Field} = \frac{\text{Force}}{\text{Source}}$$

In essence, **Sources interact with Fields and experience Forces**, and Forces are the product of an interaction between a Source and a Field.

> **i** Note
>
> **Causal direction varies by domain.** In gravitation and electrostatics, the field (g, E) exists independently of the test source — it is created by external masses or charges and can be measured without placing a test particle in it. In mechanics, "acceleration" is the *outcome* of the net force on a body, not a pre-existing environmental quantity. Writing $F = ma$ in the Source $\times$ Field form is algebraically valid but causally reversed relative to $F = qE$: force *produces* acceleration, whereas an electric field *acts on* a charge. The "Force = Source $\times$ Field" template should therefore be understood as a structural pattern whose causal interpretation varies by domain.

1.3 Intrinsic and Extrinsic

Sources carry *intrinsic* information: mass, charge, energy, even genetics. **Fields** carry *extrinsic* information about the environment. **Forces** inevitably carry information about both intrinsic and extrinsic properties of the system.

This means that it is not possible to immediately infer whether a force or outcome is due solely to intrinsic or extrinsic properties without a controlled experimental variance of either property to test for such influence.

An important caveat: the boundary between "intrinsic" and "extrinsic" is not fixed by nature — it is a modeling choice that depends on where one draws the system boundary. A quantity that is extrinsic to a subsystem (an environmental temperature, a control signal) may become intrinsic once the model is enlarged to include its source. Feedback controllers, coupled subsystems, and back-reaction effects all illustrate how the boundary shifts with the scope of analysis. We adopt the intrinsic/extrinsic decomposition throughout as a productive analytical lens, while acknowledging that it is always relative to a chosen system boundary — a point developed formally in Part XIII.

This generalization can be recognized in Lewin's equation describing the behavior of a person in their environment (Lewin 1936):

$$B = f(P, E)$$

which is generally stated as:

$$\text{Behavior} = \text{Person} \times \text{Environment}$$

The parallel here is epistemic, not algebraic — it is a recurring *pattern of confounding*, not a claim that psychology obeys the same equations as physics. Just as observing a current density $\mathbf{J}$ cannot, without auxiliary information, resolve how much is due to the conductivity σ versus the applied field $\mathbf{E}$, observing a behavior B cannot resolve how much is due to the person P versus the environment E. What recurs is the *identifiability barrier*: any interaction of two factors is structurally underdetermined from its output alone. Lewin's equation captures this epistemological point — the "Person" carries intrinsic information (traits, dispositions, genetics); the "Environment" carries extrinsic information (circumstances, context, pressures); and any attempt to attribute the observed Behavior to one or the other without controlled variation is underdetermined for the same structural reason that separating response from drive requires calibration.

1.4 The Birth of Field Thinking

It is curious to consider the possibility that Fields could be influenced by Sources themselves. Perhaps the motion of a Source through the Field has a lasting effect—like walking through snow and leaving behind a trail of deformation. The Field itself has a Source!

To explore the nature of this interaction, consider two masses placed a distance R apart. Assuming they are isolated from any other source of mass, the force between them is described by Newton's Universal Law of Gravitation:

$$F = G\frac{mM}{R^2}$$

where G is the universal gravitational constant ($G \approx 6.674 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$), M is the larger mass (the "parent" or "planet"), and m is the smaller mass (the "child" or "satellite").

To isolate the effect of the Field on one Source (say, the satellite), rearrange:

$$\frac{F}{m} = G\frac{M}{R^2}$$

Since acceleration is force per mass by Newton's Second Law:

$$g = G\frac{M}{R^2}$$

This says that the acceleration experienced by m is caused by M. From the perspective of m, the properties of the acceleration field are *extrinsic* and caused by M. The force that m experiences in the field also depends on its own *intrinsic* properties—specifically mass. The more massive m becomes, the greater the force it will experience in the same field at the same location.

The same logic applies to Coulomb's Law: $F = k\frac{qQ}{R^2}$. From the perspective of charge q, the electric field $E = k\frac{Q}{R^2}$ is extrinsic (created by Q), while the force $F = qE$ depends on both the extrinsic field and the intrinsic charge.

Since Forces are the product of Sources and Fields, they are influenced by both intrinsic and extrinsic properties.

1.5 The Universality Hint: Mass and Charge as Interchangeable

Before we proceed deeper, pause to notice a remarkable fact hidden in the equations above:

- Gravitational force depends on **mass**: $F_g = m \cdot g = m \cdot \frac{GM_s}{r^2}$
- Electric force depends on **charge**: $F_e = q \cdot E = q \cdot \frac{kQ_s}{r^2}$

The mathematical forms are **identical**. Mass plays the role of charge; G plays the role of k. They are the same equation with different labels. Yet textbooks present them as fundamentally different forces—gravity in mechanics courses, electromagnetism in separate courses. This historical accident obscures a profound unity.

Much later in this document (Part XI), we will reveal that this is no coincidence. Gravity and electricity obey the same mathematical laws up to coupling constants. When we recognize this identity, the entire framework of cross-domain analogies becomes not heuristic but necessary—a consequence of the universal structure built into the laws of physics.

For now, note the pattern: **Observable force = (intrinsic property) × (extrinsic field)**. Mass and charge both play the role of "intrinsic property." This suggests they may be more similar than different.

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2 Part II: Potential Energy and Potential

2.1 Spatial Energy

The configuration of a Source in a Field, and its corresponding Force, can also be described via its location within the Field. This description is known as **Potential Energy**—a sort of "spatial energy."

Assuming a uniform acceleration field,

Gravitational Potential Energy:

$$U = mgh = Fh$$

where the gravitational force is $F = mg$. The potential energy is a product of Force and Distance, with units of $\mathrm{N} \cdot \mathrm{m}$ = Joules. The distance h is measured from the field's zero-potential or "ground."

Electrostatic Potential Energy:

$$U = qEs = Fs$$

where the electrostatic force is $F = qE$. The potential energy is again a product of Force and Distance, also with units of Joules. The distance s is similarly measured from the fields zero-potential, which is commonly referred to a "ground".

If either a mass or a charge is "elevated" to a non-zero potential, there is energy which "wishes" to be released. Upon releasing the mass or charge, it will be driven by a force to follow a trajectory which seeks to minimize this potential energy. The force which develops as a result of this potential energy minimization can be described as the gradient of the potential function:

$$\vec{F} = -\nabla\phi$$

2.2 Potential: Energy per Unit Source

Define **potential** as the ratio of potential energy per source (specific potential energy), such that it is the product of Field and Reference Displacement.

Gravitational potential:

$$V_g = \frac{U}{m} = gh$$

Units: Joules per kilogram = $\mathrm{m}^2/\mathrm{s}^2$

Note that gravitational potential has the same units as velocity-squared—a curious fact with deep connections to energy conservation.

Electric potential (voltage):

$$V = \frac{U}{q} = Es$$

Units: Joules per Coulomb = Volts

This is the common measurement of voltage, or potential difference.

> **!** Important
>
> **Key insight:** Potential is purely *extrinsic*—it characterizes the field configuration created by external sources, independent of the test source that might be placed in it.

2.3 Work and Kinetic Energy

Moving a source through a field requires energy—specifically at the expense of potential energy. The energy involved in moving a source from one potential to another is **Work**:

$$W = Fd\cos\theta$$

where W is measured in Joules, F is force in Newtons, d is distance in meters, and θ is the angle between the applied force and the direction of motion.

More precisely:

$$W = \int \vec{F} \cdot d\vec{r}$$

Since work is the energy which contributes to change in motion along a line of action, this also defines **Kinetic Energy**:

$$W = \Delta KE$$

The change in motion is the result of net work done by the system, from both conservative forces (efficiently transferring energy) and nonconservative forces (inefficiently transferring or removing energy).

Work by conservative forces:

$$W_c = \int \vec{F}_c \cdot d\vec{r} = -\Delta PE$$

Work by nonconservative forces:

$$W_{nc} = \int \vec{F}_{nc} \cdot d\vec{r}$$

2.4 Energy Conservation

The conservation of energy statement fully describes this exchange:

$$KE_i + PE_i + W_{nc} = KE_f + PE_f$$

Rearranging:

$$W_{nc} = \Delta KE + \Delta PE$$

If all work is conservative ($W_{nc} = 0$):

$$\Delta KE + \Delta PE = 0$$

$$KE + PE = \text{constant}$$

Total mechanical energy is conserved when only conservative forces act.

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3 Part III: The Three Energy Modes

3.1 Classification of Energy Storage and Dissipation

If the energy of a system can be fully described, then its behavior can be described too. The energy of a system can be described with **three energy modes**:

- **Potential energy**: energy *stored in configuration* (position, deformation, separation)
- **Kinetic energy**: energy *stored in motion*
- **Dissipative energy**: energy *lost* to nonconservative effects (friction, drag, resistance)

Each of these three forms represents a **passive role** in the system:

1. Store energy in a potential-like way (compliance, capacitance)
2. Store energy in a kinetic-like way (inertia, inductance)
3. Dissipate energy (resistance, friction)

3.2 RLC Circuit

In a classical RLC circuit, the three passive components are the resistor, inductor, and capacitor:

- **Resistor**: the dissipative element
- **Inductor**: the kinetic storage element (in the magnetic field, and in the inertia-like behavior of current)
- **Capacitor**: the potential storage element (in the electric field, and in charge separation)

Before the circuit is closed, the capacitor is uncharged—there is no voltage across it—and the inductor carries no current. When the switch closes, current begins to flow. The resistor opposes the flow and converts electrical power into heat. The inductor opposes changes in the flow, "pushing back" against sudden jumps in current. The capacitor begins accumulating charge, increasing its voltage.

As the capacitor charges, the voltage across it rises, leaving less potential difference (voltage) across the resistor available to drive current, so current decreases. In the ideal DC limit, current goes to zero once the capacitor reaches the source voltage. The circuit evolves from kinetic activity (current) into potential storage (charge separation), while the resistor continuously bleeds energy away.

3.3 Mass-Spring-Damper

A similar behavior appears in mechanics with the mass-spring-damper system:

- **Damper**: the dissipative element—converts mechanical power into heat through friction-like effects
- **Mass**: the kinetic storage element—stores energy in motion and inertial/momentum resist changes in velocity (acceleration)
- **Spring**: the potential storage element- stores energy in deformation and "pushes back" when stretched or compressed)

Before anything moves, the spring is undeformed (no spring force) and the mass at rest (no kinetic energy). When the system is disturbed, motion begins. The damper opposes motion, producing a resistive force that scales with velocity. The mass resists rapid changes in velocity (accelerations). The spring builds a restoring force as it stores potential energy.

As spring deformation increases, the restoring force grows, reducing net force available to accelerate the mass. The velocity peaks and falls as the spring pulls the mass back toward equilibrium. The damper steadily bleeds energy, oscillations shrink, and the system settles toward rest.

This is not just a poetic similarity—the governing relationships line up in the same roles:

- The **mass** resists changes in velocity (a mechanical "inductor")
- The **spring** produces a restoring force proportional to displacement (a mechanical "capacitor")
- The **damper** produces a resistive force proportional to velocity (a mechanical "resistor")

3.4 Hydraulic Analog

A similar behavior appears in hydraulics with a restriction–inertance–accumulator system:

- **Restriction** (valve/orifice): the dissipative element—converts mechanical power into heat through viscous losses
- **Inertance** (long pipe / moving slug of fluid): the kinetic storage element—moving fluid has inertia and resists rapid changes in flow (the effect behind water hammer)
- **Compliance** (accumulator with flexible diaphragm): the potential storage element—stores energy by compressing/expanding a volume

Before anything flows, the accumulator is relaxed—no pressure difference across its diaphragm—and the fluid in the inertance pipe is stationary. When a pressure source is applied, fluid begins to move. The restriction opposes flow, converting hydraulic power into heat through viscous friction. The inertance opposes changes in flow rate, resisting sudden accelerations of the fluid column (this is the mechanism behind water hammer—fluid in motion does not stop easily).

As the accumulator fills, its diaphragm stretches and internal pressure rises, leaving less pressure difference available across the restriction to drive flow, so the flow rate decreases. In the steady-state limit, flow goes to zero once the accumulator pressure equals the source pressure. The system evolves from kinetic activity (fluid in motion) into potential storage (pressure in the accumulator), while the restriction continuously bleeds energy away.

> Note
>
> **All three systems share the same structure:** two elements exchange energy back and forth (kinetic potential), while the dissipative element continuously bleeds energy away. The nouns change—charge

and current versus displacement and velocity versus pressure and flow—but the behavior is the same:

- **Ability** — potential storage — energy latent in configuration
- **Activity** — kinetic exchange — energy in transit between forms
- **Reality** — dissipation — energy irreversibly lost

An ideal system has only Ability and Activity; a real system always includes Reality.

3.5 Effort, Flow, and the Constitutive Laws

The mapping becomes concrete when writing the passive element laws in a recognizable form. Each system has a paired set of terminal variables: an **effort** variable (the driving difference) and a **flow** variable (the through quantity). These terms originate from bond-graph theory (Paynter 1961; Karnopp, Margolis, and Rosenberg 2012), where the same three passive roles can be written consistently in effort and flow across every domain.

Electrical:

The electrical effort variable is voltage (V), and the flow variable is current (I). The three passive circuit elements—resistor, inductor, and capacitor—each relate these two variables through a distinct constitutive law: proportionality, differentiation, or integration.

- **Resistor:**

  $$V_R = RI$$

  where R is resistance, measured in Ohms (Ω). Voltage drop is proportional to current. The faster charge flows, the more energy the resistor bleeds as heat.

- **Inductor:**

  $$V_L = L\frac{dI}{dt}$$

  where L is inductance, measured in Henrys (H). Voltage is proportional to the *rate of change* of current, not the quantity of current itself. The inductor stores energy in a magnetic field while current builds, then returns it as current collapses.

- **Capacitor:**

  $$I_C = C\frac{dV}{dt}$$

  where C is capacitance, measured in Farads (F). Current flows into the capacitor only while its voltage is changing. Once fully charged, current stops. Energy is stored in the electric field between the plates, held in the separation of charge.

Gravitational:

The gravitational effort variable is potential (V_g), and the flow variable is mass flow rate ($\dot{m}$). Gravitational potential is a true potential — energy per unit mass (J/kg) — making it the closest mechanical analog to voltage and pressure. The same three constitutive forms apply:

- **Gravitational resistance** (drag / viscous friction on mass flow):

  $$V_g = R_g\dot{m}$$

  where R_g is gravitational resistance, measured in $\mathrm{J \cdot s/kg^2}$. Potential drop is proportional to mass flow rate. The faster mass flows through the resistive path, the more energy is lost.

- **Gravitational inertance** (resistance to changes in mass flow):

  $$V_g = L_g\frac{d\dot{m}}{dt}$$

  where L_g is gravitational inertance, measured in $\mathrm{J \cdot s^2/kg^2}$. Potential is proportional to the *rate of change* of mass flow, not mass flow itself.

- **Gravitational compliance** (mass storage under potential):

$$\dot{m} = H\frac{dV_g}{dt}$$

  where H is gravitational compliance, measured in $\mathrm{kg}^2/(\mathrm{J}\cdot\mathrm{s})$. Mass flow into storage is proportional to the rate of change of potential.

Mechanical (translation):

The mechanical effort variable is force (F), and the flow variable is velocity ($v = \dot{x}$). The three passive mechanical elements—damper, mass, and spring—each relate these two variables through the same three constitutive forms: proportionality, differentiation, or integration. Note that unlike voltage, pressure, and gravitational potential, force is not a traditional potential — it is the gradient of potential energy with respect to displacement ($F = -dU/dx$). Nevertheless, force carries units of energy per displacement (J/m = N), and the effort-flow structure holds. The familiar force laws ($F = bv$, $F = ma$, $F = kx$) can be rewritten in effort-flow form: Newton's second law becomes $F = m\frac{dv}{dt}$ since $a = \frac{dv}{dt}$, and Hooke's law becomes $v = \frac{1}{k}\frac{dF}{dt}$ after differentiating both sides, where the compliance $\frac{1}{k}$ plays the role of capacitance.

- **Damper:**

$$F_d = bv$$

  where b is the damping coefficient, measured in Newton-seconds per meter ($\mathrm{N}\cdot\mathrm{s/m}$). Resistive force is proportional to velocity. The faster the mass moves, the harder the damper pushes back, converting kinetic energy to heat.

- **Mass:**

$$F_m = m\frac{dv}{dt}$$

  where m is mass, measured in kilograms (kg). Force is proportional to the *rate of change* of velocity, not velocity itself. The mass does not resist motion—it resists *changes* in motion (accelerations). A moving mass carries momentum past equilibrium, overshooting the spring's rest position and enabling oscillation.

- **Spring:**

$$v = \frac{1}{k}\frac{dF_s}{dt}$$

  where k is the spring stiffness, measured in Newtons per meter (N/m), and $\frac{1}{k}$ is the mechanical compliance. Velocity is proportional to the rate of change of the restoring force. A softer spring (larger $\frac{1}{k}$) allows more velocity for the same rate of force change.

> **i** Note
>
> The reader can verify that **mechanical rotation** follows the same pattern with effort = torque τ (J/rad) and flow = angular velocity ω (rad/s), yielding the constitutive laws $\tau = b_\theta\omega$, $\tau = J\frac{d\omega}{dt}$, and $\omega = \frac{1}{k_\theta}\frac{d\tau}{dt}$.

Hydraulic:

The hydraulic effort variable is pressure (p), and the flow variable is volume flow rate (Q). The three passive hydraulic elements—restriction, inertance, and compliance—each relate these two variables through the same three constitutive forms: proportionality, differentiation, or integration. The inertance of a fluid column is $I_h = \frac{\rho L}{A}$ (where ρ is density, L is pipe length, and A is cross-sectional area), and the compliance of an accumulator is C_h, relating stored volume to pressure via $V_{\text{stored}} = C_h\, p$.

- **Restriction** (valve/orifice):

$$p = R_h Q$$

where R_h is hydraulic resistance, measured in Pascal-seconds per cubic meter ($\mathrm{Pa} \cdot \mathrm{s/m^3}$). Pressure drop is proportional to flow rate. The faster fluid moves through the restriction, the more energy is lost to viscous heating.

- **Inertance** (long pipe / fluid slug):

$$p = I_h \frac{dQ}{dt}$$

  where I_h is inertance, measured in Pascal-seconds-squared per cubic meter ($\mathrm{Pa} \cdot \mathrm{s^2/m^3}$). Pressure is proportional to the *rate of change* of flow, not flow itself. A long, narrow pipe resists rapid changes in flow rate—this is the mechanism behind water hammer.

- **Compliance** (accumulator with flexible diaphragm):

$$Q = C_h \frac{dp}{dt}$$

  where C_h is hydraulic compliance, measured in cubic meters per Pascal ($\mathrm{m^3/Pa}$). Flow into the accumulator is proportional to the rate of change of pressure. A more compliant accumulator accepts more flow for the same rate of pressure rise.

The structural correspondence across all domains is summarized below.

Role

Electrical

Gravitational

Mechanical

Hydraulic

Pattern

translation

rotation

Dissipation

$V = RI$

$V_g = R_g \dot{m}$

$F = bv$

$\tau = b_\theta \omega$

$p = R_h Q$

effort = resistance × flow

Kinetic storage

$V = L\frac{dI}{dt}$

$V_g = L_g \frac{d\dot{m}}{dt}$

$F = m\frac{dv}{dt}$

$\tau = J\frac{d\omega}{dt}$

$p = I_h \frac{dQ}{dt}$

effort = inertance × $\frac{d}{dt}$(flow)

Potential storage

$I = C\frac{dV}{dt}$

$\dot{m} = H\frac{dV_g}{dt}$

$v = \frac{1}{k}\frac{dF}{dt}$

$\omega = \frac{1}{k_\theta}\frac{d\tau}{dt}$

$Q = C_h\frac{dp}{dt}$

flow = compliance $\times$ $\frac{d}{dt}$(effort)

The structural pattern is clear. Each row of the table compares a single universal statement:

- **Reality** (dissipation): the dissipative element produces a response proportional to the flow variable — an instantaneous, memoryless tax on any real process.
- **Activity** (kinetic storage): the kinetic storage element resists changes in the flow variable — energy carried in the momentum of the flow itself.
- **Ability** (potential storage): the flow variable equals a compliance times the rate of change of the effort variable — energy held latent in configuration.

In every case, the same calculus applies — proportionality, differentiation, integration — only the names and variables differ.

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4 Part IV: Impedance, Power, and Bond Graphs

4.1 Terminal Pairs

The constitutive laws were written in effort-flow form, but the effort and flow variables were described by domain — voltage and current, force and velocity, etc,. These variables have general dimensions. In every domain, the effort variable carries units of energy per source, and the flow variable carries units of source per time. Making this explicit reveals why impedance and power take the same form everywhere.

The effort variable carries units of **energy per source**. In most domains, this is a true potential:

- Electrical: voltage V — energy per unit charge (J/C)
- Gravitational: potential V_g — energy per unit mass (J/kg)
- Hydraulic: pressure p — energy per unit volume (J/m^3)

In mechanical systems, the effort variable has the same dimensional structure but is not traditionally called a potential:

- Mechanical (translation): force F — energy per unit displacement (J/m = N)
- Mechanical (rotation): torque τ — energy per unit angle (J/rad)

Force and torque are gradients of potential energy with respect to their source coordinates ($F = -dU/dx$, $\tau = -dU/d\theta$), not potentials themselves. The dimensional pattern — energy per source — holds in every case, but the naming convention diverges for mechanical systems.

The flow variable carries units of **source per time**:

- Electrical: current I — charge per time (C/s = A)
- Gravitational: mass flow rate $\dot{m}$ — mass per time (kg/s)
- Hydraulic: volume flow rate Q — volume per time (m^3/s)
- Mechanical (translation): velocity v — displacement per time (m/s)
- Mechanical (rotation): angular velocity ω — angle per time (rad/s)

The framework does not prescribe what the source must be; it requires only that the effort-flow pairing be power-conjugate — that their product yield power. Different choices of "source" within the same physical domain (e.g., displacement vs. mass for mechanical systems) yield different but equally valid descriptions.

These two generalized dimensions — effort and flow — combine to produce two fundamental system quantities through division and multiplication.

Their ratio is impedance. Dividing effort by flow:

$$Z = \frac{\text{effort}}{\text{flow}} = \frac{\text{energy/source}}{\text{source/time}} = \frac{\text{energy} \cdot \text{time}}{\text{source}^2}$$

Impedance encodes the dissipative and storage qualities of the system. It controls how much driving difference is required per unit of through quantity—how quickly energy is bled away relative to how much is stored:

- Electrical: $Z = \frac{V}{I} = \frac{\text{J/C}}{\text{C/s}} = \frac{\text{J·s}}{\text{C}^2}$ (energy · time / charge^2)
- Gravitational: $Z = \frac{V_g}{\dot{m}} = \frac{\text{J/kg}}{\text{kg/s}} = \frac{\text{J·s}}{\text{kg}^2}$ (energy · time / mass^2)
- Mechanical (translation): $Z = \frac{F}{v} = \frac{\text{J/m}}{\text{m/s}} = \frac{\text{J·s}}{\text{m}^2}$ (energy · time / displacement^2)
- Mechanical (rotation): $Z = \frac{\tau}{\omega} = \frac{\text{J/rad}}{\text{rad/s}} = \frac{\text{J·s}}{\text{rad}^2}$ (energy · time / angle^2)
- Hydraulic: $Z = \frac{p}{Q} = \frac{\text{J/m}^3}{\text{m}^3\text{/s}} = \frac{\text{J·s}}{\text{m}^6}$ (energy · time / volume^2)

Every domain reduces to the same generalized form — the "source" differs (charge, displacement, angle, mass, volume), but the dimensional structure is universal. Note that hydraulic pressure p denotes a differential pressure — the drop across the element measured with respect to an absolute or gauge reference, analogous to voltage measured with respect to ground.

Their product is power. Multiplying effort by flow:

$$P = \text{effort} \times \text{flow} = \frac{\text{energy}}{\text{source}} \times \frac{\text{source}}{\text{time}} = \frac{\text{energy}}{\text{time}}$$

The source units cancel, leaving power as the rate of energy transfer—regardless of domain:

- Electrical: $P = VI$ — measured in Watts (W = J/s)
- Gravitational: $P = V_g\dot{m}$ — measured in Watts
- Mechanical (translation): $P = Fv$ — measured in Watts
- Mechanical (rotation): $P = \tau\omega$ — measured in Watts
- Hydraulic: $P = p \cdot Q$ — measured in Watts

Describing a system at its terminals using (V, I) or (F, v) or $(V_g, \dot{m})$ or (p, Q) makes it possible to directly track **energy transfer**.

> Tip
>
> **Why real sources "sag under load."** Real sources contain internal dissipative effects, so part of the effort is spent internally when the flow increases. A battery behaves like an EMF in series with internal resistance: $V_{\text{terminal}} = \mathcal{E} - rI$. A pressurized canister behaves identically in hydraulics: reservoir pressure is the available effort, but outlet pressure falls as flow increases because pressure is lost across internal restrictions. Same structure, different words and variables.

4.2 The Bridge to Dynamical Systems

The moment a system includes storage — something that accumulates over time — its behavior stops being a purely algebraic ratio at the terminal and becomes an evolution law. Impedance is no longer a static number; it becomes a dynamic, frequency-dependent quantity because part of the effort is temporarily held in storage before being returned or dissipated. That is where ordinary differential equations (and, when storage is distributed in space, partial differential equations) enter the conversation.

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5 Part V: The Complete Cross-Domain Analogy

Before developing the mathematical framework promised above, it is worth consolidating everything established so far — sources, displacements, effort-flow pairs, power conjugacy, impedance, and storage elements — into a single reference. The following table organizes the cross-domain analogies and will serve as the dictionary for the formal treatment that follows in Part VI:

Component / Parameter	Gravitational	Electrical	Hydraulic	Mechanical (trans.)	Mechanical (rot.)
Source	Mass m (kg)	Charge q (C)	Volume V (m^3)	Displacement x (m)	Angle θ (rad)
Displacement	Height h	Path coordinate x	Volume V	Position x	Angle θ
Motion	Mass flow $\dot{m}$	Drift speed u_d	Volume flow $Q = \dot{V}$	Velocity $v = \dot{x}$	Angular speed $\omega = \dot{\theta}$
Field	Gravitic field g	Electric field E	Pressure gradient $-\nabla p$	Acceleration a	Angular accel. α
Force = Source · Field	$F = mg$	$F = qE$	$F = pA$	$F = ma^\dagger$	$\tau = J\alpha^\dagger$
Work-Energy	$U = \int mg\,dh$	$U = \int V\,dq$	$U = \int p\,dV$	$U = \int F\,dx$	$U = \int \tau\,d\theta$
Potential (energy/source)	Grav. potential $V_g = U/m = gh$ (J/kg)	Voltage $V = U/q$ (J/C)	Pressure $p = U/V$ ($\mathrm{J/m^3}$)	Force F (J/m)	Torque τ (J/rad)
Flux / Current (source/time)	Mass flow $\dot{m}$ (kg/s)	$I = \dot{q}$ (C/s)	$Q = \dot{V}$ ($\mathrm{m^3/s}$)	$v = \dot{x}$ (m/s)	$\omega = \dot{\theta}$ (rad/s)
Power	$P = V_g\dot{m}$	$P = VI$	$P = p \cdot Q$	$P = Fv$	$P = \tau\omega$
Impedance	$Z_g = V_g/\dot{m}$	$Z = V/I$	$Z = p/Q$	$Z = F/v$	$Z = \tau/\omega$
Resistance (R-type)	$V_g = R_g\dot{m}$ Grav. resistance R_g	$V = RI$ Resistance R	$p = R_hQ$ Hyd. resistance R_h	$F = bv$ Damping b	$\tau = b_\theta\omega$ Rot. damping b_θ
Conductance (G-type)	$G_g = 1/R_g$	$G = 1/R$	$G_h = 1/R_h$	$\mu = 1/b$	$\mu_\theta = 1/b_\theta$
Capacitance (C-type)	$m = HV_g$ Grav. capacitance H	$q = CV$ Capacitance C	$V_{\text{stored}} = C_hp$ Hyd. capacitance C_h	$x = C_mF$ Compliance $C_m = 1/k$	$\theta = C_\theta\tau$ Rot. compliance $C_\theta = 1/k_\theta$
Stiffness (inverse C)	$1/H$	$1/C$	$E_h = 1/C_h$	k Spring constant	k_θ
Inductance (I-type)	$V_g = L_g\frac{d\dot{m}}{dt}$ Grav. inductance L_g	$V = L\dot{I}$ Inductance L	$p = I_h\dot{Q}$ Inertance I_h	$F = m\dot{v}$ Mass m	$\tau = J\dot{\omega}$ Moment of inertia J
C-state ($\int f\,dt$)	Mass m	Charge q	Volume V	Displacement x	Angle θ
I-state ($\int e\,dt$)	Potential-impulse π_g	Flux linkage λ	Pressure impulse Π	Momentum p	Angular mom. L

Note

The **C-state** and **I-state** rows introduce the generalized displacement and momentum variables used in bond-graph and Hamiltonian formulations. These have not yet been discussed but are included here for completeness; their meaning and significance will become clear in the mathematical framework that follows.

† In the mechanical columns, acceleration a (and angular acceleration α) is not an independent environmental field like g or E; it is the kinematic result of the net force. The Source · Field form is used here for structural consistency across domains, but the causal arrow is reversed — see the note in Part I.

The cross-domain analogies in this table are not merely pedagogical heuristics; Part XII proves formally that they are mathematical necessities — consequences of power conjugacy and conservation laws.

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6 Part VI: Mathematical Framework

6.1 Finite-Dimensional Dynamical Systems

Start with a forced mass-spring-damper system:

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

where $x(t)$ is position—an unknown function of time—m, b, k are intrinsic parameters of the system (mass, damping, and stiffness), and $F(t)$ is an extrinsic time-dependent forcing function.

6.1.1 State-Space Formulation

To express this forced harmonic oscillator as a standard first-order dynamical system, define the state vector:

$$z_1 = x, \qquad z_2 = \dot{x}, \qquad \mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

Taking derivatives:

$$\dot{z}_1 = \dot{x} = z_2, \qquad \dot{z}_2 = \ddot{x} = \frac{1}{m}\left(F(t) - bz_2 - kz_1\right)$$

As a system:

$$\dot{\mathbf{z}} = \begin{pmatrix} \dot{z}_1 \\ \dot{z}_2 \end{pmatrix} = \begin{pmatrix} z_2 \\ \frac{1}{m}F(t) - \frac{k}{m}z_1 - \frac{b}{m}z_2 \end{pmatrix}$$

Defining the intrinsic parameters as a vector $\theta = (m, b, k)$, the input driving force as $u(t) = F(t)$, and the state vector as $\mathbf{z} = (z_1, z_2)$, and packing these all into one function:

$$\dot{\mathbf{z}} = f(\mathbf{z}, t; \theta, u(t)) = \begin{pmatrix} z_2 \\ -\frac{k}{m}z_1 - \frac{b}{m}z_2 + \frac{1}{m}u(t) \end{pmatrix}$$

Let the state be a point in the position-velocity plane, $\mathbf{z}(t) \in \mathbb{R}^2$. Separating the intrinsic linear component from the extrinsic forcing component:

$$\dot{\mathbf{z}}(t) = \underbrace{\begin{pmatrix} 0 & 1 \\ -k/m & -b/m \end{pmatrix}}_{\mathbf{A}_\theta} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ F(t)/m \end{pmatrix}}_{\mathbf{s}(t)}$$

The **intrinsic matrix $\mathbf{A}_\theta$** encodes the system's own dynamics through its parameters, and $\mathbf{s}(t)$ is the **extrinsic forcing vector**. In reduced form:

$$\dot{\mathbf{z}}(t) = \mathbf{A}_\theta \mathbf{z}(t) + \mathbf{s}(t)$$

6.2 Chains of Coupled Oscillators

To see how this structure scales beyond a single degree of freedom, consider a chain of n masses connected by springs and dampers, with positions $x_i(t)$ and external forces $F_i(t)$ for each mass i.

6.2.1 Matrix Formulation

Define the state vector by stacking all positions followed by all velocities:

$$\mathbf{z} = \begin{pmatrix} \mathbf{x} \\ \dot{\mathbf{x}} \end{pmatrix} = [x_1, \dots, x_n, \dot{x}_1, \dots, \dot{x}_n]^T \in \mathbb{R}^{2n}$$

The intrinsic matrix takes the block form:

$$\mathbf{A}_\theta = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{pmatrix}$$

with $\mathbf{0}$ and $\mathbf{I}$ being $n \times n$ zero and identity matrices, $\mathbf{M}$ a diagonal mass matrix, and $\mathbf{K}$ and $\mathbf{C}$ the stiffness and damping matrices respectively.

The **mass matrix** is diagonal:

$$\mathbf{M} = \begin{pmatrix} m_1 & 0 & \cdots & 0 \\ 0 & m_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_n \end{pmatrix}$$

For identical masses, this reduces to $\mathbf{M} = m\mathbf{I}$.

The **stiffness matrix $\mathbf{K} \in \mathbb{R}^{n \times n}$**, assuming each mass is connected to its neighbors by springs of stiffness k_i and that the first and last masses are connected to fixed walls (also by springs), is tridiagonal:

$$\mathbf{K} = \begin{pmatrix} k_0 + k_1 & -k_1 & 0 & \cdots & 0 \\ -k_1 & k_1 + k_2 & -k_2 & \ddots & \vdots \\ 0 & -k_2 & k_2 + k_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -k_{n-1} \\ 0 & \cdots & 0 & -k_{n-1} & k_{n-1} + k_n \end{pmatrix}$$

In index form:

- $K_{ii} = k_{i-1} + k_i, \;\; i = 1, \dots, n$
- $K_{i,i+1} = K_{i+1,i} = -k_i, \;\; i = 1, \dots, n-1$
- All other $K_{ij} = 0$

Were the masses influenced by the next-nearest neighbor, the matrix would be pentadiagonal (springs between mass i and $i+2$). The offset of a nonzero diagonal represents the proximity of physical coupling between masses.

For identical springs $k_i = k$, each diagonal entry becomes $2k$ and each off-diagonal becomes $-k$.

The **damping matrix** $\mathbf{C} \in \mathbb{R}^{n\times n}$ has the same tridiagonal pattern with b:

$$\mathbf{C} = \begin{pmatrix} b_0 + b_1 & -b_1 & 0 & \cdots & 0 \\ -b_1 & b_1 + b_2 & -b_2 & \ddots & \vdots \\ 0 & -b_2 & b_2 + b_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -b_{n-1} \\ 0 & \cdots & 0 & -b_{n-1} & b_{n-1} + b_n \end{pmatrix}$$

In index form:

- $C_{ii} = b_{i-1} + b_i,\ \ i = 1, \dots, n$
- $C_{i,i+1} = C_{i+1,i} = -b_i,\ \ i = 1, \dots, n-1$
- All other $C_{ij} = 0$

For identical dampers $b_i = b$, this simplifies analogously.

The extrinsic forcing vector collects all external driving forces:

$$\mathbf{s}(t) = \begin{pmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{F}(t) \end{pmatrix}, \qquad \mathbf{F}(t) = [F_1(t), \dots, F_n(t)]^T$$

6.2.2 Physical Interpretation

Now assume identical masses m, identical springs of stiffness k, and identical dashpots of damping b. For an interior mass i, the net spring force is $k(x_{i-1} - 2x_i + x_{i+1})$ and the net damping force is $b(\dot{x}_{i-1} - 2\dot{x}_i + \dot{x}_{i+1})$. Newton's second law gives:

$$m\ddot{x}_i = k(x_{i-1} - 2x_i + x_{i+1}) + b(\dot{x}_{i-1} - 2\dot{x}_i + \dot{x}_{i+1}) + F_i(t)$$

The coefficient pattern $(1, -2, 1)$ is the **discrete second derivative stencil**—exactly the tridiagonal pattern encoded in $\mathbf{K}$ and $\mathbf{C}$. The system again takes the form:

$$\dot{\mathbf{z}}(t) = \mathbf{A}_\theta \mathbf{z}(t) + \mathbf{s}(t)$$

6.3 The Continuum Limit

The nearest-neighbor chain already contains the spatial structure of a one-dimensional elastic medium. The $(1, -2, 1)$ stencil in the discrete equation is a finite-difference approximation to a second spatial derivative—so as the number of masses grows and the spacing shrinks, the chain should converge to a continuous wave equation.

To pass to a continuum, introduce a uniform spacing a between masses and identify mass i with position $x_i = i \cdot a$ along a one-dimensional body. Approximate the discrete displacements by a smooth field $u(x,t)$ via $x_i(t) \approx u(x_i, t)$, and relate the lumped parameters to continuum quantities:

- Mass per unit length (ρ density, A cross-section): $m = \rho A\, a$
- Axial stiffness (E Young's modulus): $k = EA/a$
- Damping (η viscous coefficient): $b = \eta A/a$
- Force per unit length: $f(x_i, t) \approx F_i(t)/a$

Substituting these into the discrete equation of motion and dividing by a:

$$\rho A \left.\frac{\partial^2 u}{\partial t^2}\right|_{x_i} = EA\frac{u(x_{i+1},t) - 2u(x_i,t) + u(x_{i-1},t)}{a^2} + \eta A\frac{\dot{u}(x_{i+1},t) - 2\dot{u}(x_i,t) + \dot{u}(x_{i-1},t)}{a^2} + f(x_i,t)$$

The finite difference quotients

$$\frac{u(x_{i+1},t) - 2u(x_i,t) + u(x_{i-1},t)}{a^2} \longrightarrow \left.\frac{\partial^2 u}{\partial x^2}\right|_{x_i}$$

(and the same for $\dot{u}$) are central second differences, which converge to second spatial derivatives as $a \to 0$. In the continuum limit:

$$\rho A\frac{\partial^2 u}{\partial t^2} = EA\frac{\partial^2 u}{\partial x^2} + \eta A\frac{\partial^3 u}{\partial x^2 \partial t} + f(x,t)$$

Dividing by ρA and defining:

- Wave speed: $c^2 = E/\rho$
- Damping parameter: $\gamma = \eta/\rho$
- Body-force term: $g(x,t) = f(x,t)/(\rho A)$

yields the **damped wave equation** of Kelvin-Voigt type (damping proportional to strain rate $\partial^2 u/\partial x \partial t$):

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} + \gamma\frac{\partial^3 u}{\partial x^2 \partial t} + g(x,t)$$

Setting $\gamma = 0$ yields the **undamped wave equation**:

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} + g(x,t)$$

with boundary conditions $u(0,t) = u(L,t) = 0$ for a string of length $L = na$ with fixed ends.

The Kelvin-Voigt damping term $\gamma\, \partial^3 u/\partial x^2 \partial t$ arose because we placed dashpots *between* neighboring masses—dissipation through spatial gradients in velocity. An alternative is **Rayleigh damping**: if each mass instead has a dashpot to ground (force $-b\dot{x}_i$ rather than differences), the continuum damping term becomes $\gamma\, \partial u/\partial t$ rather than $\gamma\, \partial^3 u/\partial x^2 \partial t$, yielding:

$$\frac{\partial^2 u}{\partial t^2} + \gamma\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2} + g(x,t)$$

With free ends, the boundary conditions become $\partial u/\partial x|_{x=0} = \partial u/\partial x|_{x=L} = 0$. The choice of damping model and boundary conditions reflects intrinsic structural decisions about the physical system.

6.4 Field Evolution as Operator Equation

Introduce the first-order state field $\mathbf{w}(x,t) = (u(x,t),\ v(x,t))^T$ where $v(x,t) = \partial u/\partial t$. Then the wave equation becomes the first-order system:

$$\frac{\partial u}{\partial t} = v, \qquad \frac{\partial v}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2} + \gamma\frac{\partial^3 u}{\partial x^2 \partial t} + g(x,t)$$

Setting $\gamma = 0$ for a clean display, this can be written in matrix form:

$$
\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ c^2\frac{\partial^2}{\partial x^2} & 0 \end{pmatrix}}_{\mathcal{L}_\theta}\begin{pmatrix} u \\ v \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ g(x,t) \end{pmatrix}}_{\mathbf{S}(x,t)}
$$

In reduced form:

$$
\frac{\partial \mathbf{w}}{\partial t} = \mathcal{L}_\theta[\mathbf{w}] + \mathbf{S}(x,t)
$$

The **intrinsic operator** $\mathcal{L}_\theta$ is determined by:

- Material parameters (E, ρ, η),
- Geometry (string length, boundary conditions),
- The choice of damping model (which would add terms to the lower-left and lower-right blocks).

The **extrinsic source** $\mathbf{S}(x,t)$ encodes distributed forcing along the string. This is exactly the field-level analogue of the finite chain equation $\dot{\mathbf{z}} = \mathbf{A}_\theta \mathbf{z} + \mathbf{s}(t)$, with $\mathbf{A}_\theta$ replaced by $\mathcal{L}_\theta$ and $\mathbf{s}(t)$ replaced by $\mathbf{S}(x,t)$.

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7 Part VII: Conservation Laws and Constitutive Relations

7.1 The Generic Pattern

Many field equations arise from combining:

1. **Conservation law:** $\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{J} = S$
2. **Constitutive relation:** $\mathbf{J} = \text{Response} \times \text{Drive}$ (linear case; more generally, $\mathbf{J} = \mathcal{R}[\text{Drive}]$)

7.2 Heat Conduction

Conservation of energy:

$$
\rho c_p \frac{\partial T}{\partial t} + \nabla \cdot \mathbf{q} = Q
$$

Fourier's Law:

$$
\mathbf{q} = -k\nabla T
$$

Heat equation:

$$
\frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{Q}{\rho c_p}
$$

where $\alpha = k/(\rho c_p)$ is **thermal diffusivity** [m²/s].

Intrinsic: α (thermal diffusivity) **Extrinsic:** Q (heat source), boundary temperatures

7.3 Mass Diffusion

Conservation of mass:

$$
\frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J} = S
$$

Fick's Law:

$$
\mathbf{J} = -D\nabla c
$$

Diffusion equation:

$$
\frac{\partial c}{\partial t} = D\nabla^2 c + S
$$

7.4 Charge Transport

Conservation of charge:

$$\frac{\partial \rho_e}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

Ohm's Law:

$$\mathbf{J} = \sigma \mathbf{E} = -\sigma \nabla V$$

7.5 Linear Elasticity

Conservation of momentum:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \sigma + \mathbf{f}$$

Hooke's Law:

$$\sigma = \mathbf{C} : \varepsilon$$

where $\mathbf{C}$ is the fourth-order stiffness tensor and $\varepsilon = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$.

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8 Part VIII: Manifolds and Trajectories — Rigorous Formulation

8.1 PDEs as Infinite-Dimensional ODEs

A PDE like $\frac{\partial u}{\partial t} = \mathcal{L}_\theta[u] + S$ is **rigorously equivalent** to an ODE on an infinite-dimensional Hilbert space (Pazy 1983). At each instant t, the state is not a finite vector in $\mathbb{R}^n$ but an entire field configuration $u(\cdot, t)$—a point in an infinite-dimensional function space.

Formal Statement: Let $H = L^2(\Omega)$ be the Hilbert space of square-integrable functions on spatial domain Ω. A PDE of the form

$$\frac{\partial u}{\partial t} = \mathcal{L}_\theta[u] + S(x, t)$$

with boundary conditions can be written **rigorously** as the abstract evolution equation

$$\frac{d\mathbf{u}}{dt} = \mathbf{A}\mathbf{u} + \mathbf{S}(t), \quad \mathbf{u}(0) = \mathbf{u}_0$$

where: - $\mathbf{u}(t) \in H$ is the state (field configuration at time t) - $\mathbf{A} : D(\mathbf{A}) \subset H \to H$ is a densely-defined linear operator with domain $D(\mathbf{A})$ - $\mathbf{S}(t) \in H$ is the external forcing (extrinsic source)

The functional-analytic machinery is developed in Appendix N.

8.1.1 The Operator Determines the Geometry

The intrinsic operator $\mathcal{L}_\theta$ (equivalently, $\mathbf{A}$) determines the **geometry** of the infinite-dimensional state space:

- **Eigenfunctions (Modes):** Solutions ϕ_n to the eigenvalue problem $\mathbf{A}\phi_n = \lambda_n \phi_n$ represent the natural shapes the system likes to assume. They form an orthonormal basis for H (under suitable conditions).
- **Spectrum (Eigenvalues):** The set $\sigma(\mathbf{A}) = \{\lambda_n : \mathbf{A}\phi_n = \lambda_n \phi_n\}$ determines the long-time behavior of perturbations:
  - If $\mathrm{Re}(\lambda_n) < 0$, mode ϕ_n **decays exponentially** at rate $e^{\mathrm{Re}(\lambda_n)t}$
  - If $\mathrm{Re}(\lambda_n) > 0$, mode ϕ_n **grows exponentially** at rate $e^{\mathrm{Re}(\lambda_n)t}$

- If $\mathrm{Im}(\lambda_n) \neq 0$, mode ϕ_n **oscillates** at frequency $|\mathrm{Im}(\lambda_n)|$

- **Stability:** The entire infinite-dimensional system is **globally stable** iff $\sup_n \mathrm{Re}(\lambda_n) < 0$ (all eigenvalues have negative real part).

This geometric structure exists independent of any particular trajectory—it is built into the intrinsic properties (material parameters, domain shape, boundary conditions).

8.1.2 Example: Heat Equation on $[0,1]$

The heat equation $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$ on the interval $[0,1]$ with Dirichlet boundary conditions ($u(0,t) = u(1,t) = 0$) has:

- **Eigenfunctions (Modes):** $\phi_n(x) = \sin(n\pi x)$ for $n = 1, 2, 3, ...$
- **Eigenvalues (Decay Rates):** $\lambda_n = -\alpha(n\pi)^2$ for each mode n
- **Geometry:** The state space $H = L^2([0,1])$ is decomposed into infinite-dimensional subspaces, each labeled by n, with mode n decaying at rate $\alpha n^2 \pi^2$

High-frequency modes (large n) decay rapidly; low-frequency modes (small n) decay slowly. This is why the heat equation smooths out sharp features over time—it kills the high-frequency content.

8.2 Trajectories as Realized Histories

Extrinsic drives and initial conditions, by contrast, select which trajectory is realized—which path through the infinite-dimensional manifold actually occurs:

- **The forcing $\mathbf{S}(t) \in H$** injects energy into particular modes (directions in the infinite-dimensional space), pushing the system away from equilibrium in those directions.

- **Initial conditions $\mathbf{u}_0 \in H$** determine the starting point on the manifold. The Fourier expansion $\mathbf{u}_0 = \sum_n c_n \phi_n$ shows how much weight the initial condition places in each mode.

- **Boundary conditions** constrain which modes can be active and restrict the accessible region of the manifold. Dirichlet boundaries force $u = 0$ at the boundary; Neumann boundaries allow $\partial u / \partial n = 0$, affecting which eigenfunctions are admissible.

In summary: the operator $\mathbf{A}$ defines the geometry (modes, decay rates, stability); the trajectory $\mathbf{u}(t)$ is uniquely determined by $\mathbf{A}$, initial data $\mathbf{u}_0$, and forcing $\mathbf{S}(t)$.

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9 Part IX: Causal Attribution and Product Structure

9.1 The Fundamental Problem

The simple algebraic relation

$$B = R \times D$$

already illustrates a key epistemic issue. Given only B (behavior), there is no unique way to factor it into R (response) and D (drive).

For any nonzero scalar α:

$$B = R \times D = (\alpha R) \times \left(\frac{D}{\alpha}\right)$$

represents the same behavior. Without additional information, one cannot say how much of B is due to R versus D.

9.2 Physical Interpretation

Consider $\mathbf{J} = \sigma\mathbf{E}$ (Ohm's law). Observing current density $\mathbf{J}$, we cannot distinguish:

- High conductivity σ with weak field $\mathbf{E}$
- Low conductivity σ with strong field $\mathbf{E}$
- Any intermediate combination

Without independent measurement of either σ or $\mathbf{E}$, the attribution is underdetermined.

9.3 The Matrix Case

For vector systems $\mathbf{J} = \mathbf{L} \cdot \mathbf{X}$:

$$\mathbf{J} = \mathbf{L} \cdot \mathbf{X} = (\mathbf{L} \cdot \mathbf{M}) \cdot (\mathbf{M}^{-1} \cdot \mathbf{X})$$

for any invertible matrix $\mathbf{M}$. The pair $(\mathbf{L} \cdot \mathbf{M}, \mathbf{M}^{-1} \cdot \mathbf{X})$ produces the same flux $\mathbf{J}$.

9.4 Resolving Ambiguity

To separate intrinsic from extrinsic contributions, one must:

1. **Control the drive:** Apply known, calibrated forces/fields
2. **Hold intrinsic properties fixed:** Same material, temperature, geometry
3. **Independent measurement:** Measure response coefficients separately

This is the logic of experimental calibration and material characterization.

9.5 The Nature-Nurture Parallel

The same product structure appears in discussions of nature versus nurture:

$$\text{Phenotype} = \text{Genotype} \times \text{Environment}$$

Observed phenotypes cannot uniquely reveal genetic versus environmental contributions without controlled comparisons (twin studies, common-garden experiments).

9.6 Epistemological Implications

The product structure implies:

- Behavior alone **cannot** support sharp causal attribution
- Auxiliary controls and measurements are **always** required
- Claims of "pure" intrinsic or extrinsic causation are **structurally unjustified**

This is not a limitation of measurement precision—it is a mathematical feature of product-structured models. These epistemological barriers are formalized in Part XV, which develops identifiability theorems showing that certain causal attributions remain underdetermined even with unlimited data.

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